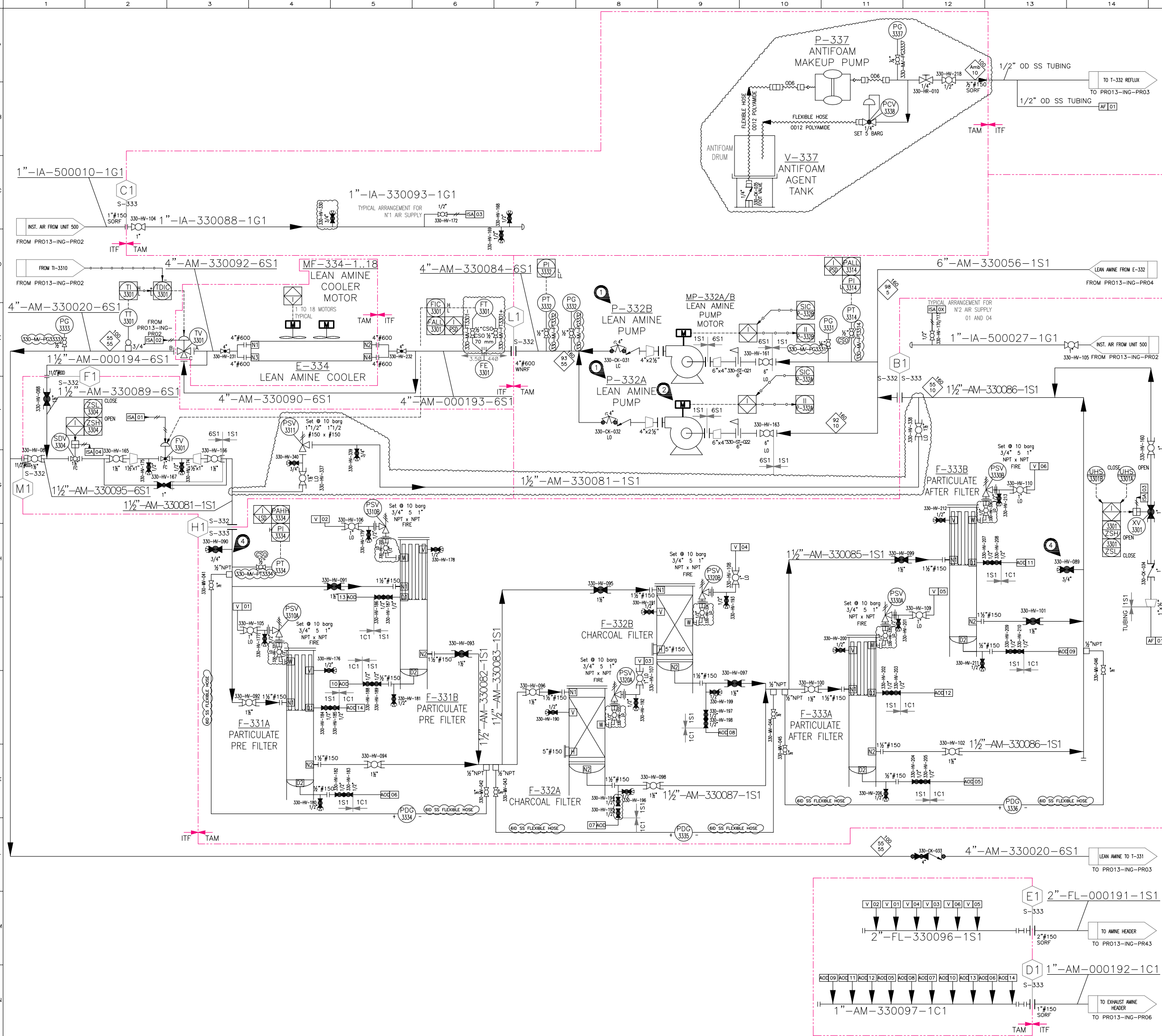




REFERENCE DOCUMENTS	
P&ID LEGEND	PRO13-ING-PRO0
PROCESS FLOW DIAGRAM	21044-PR-PID-00000
UTILITY FLOW DIAGRAM	PRO10-ING-PRO1/02/03/04
MATERIAL SELECTION DIAGRAM	21044-PR-PFD-00001
	PRO10-ING-PR11
	21044-PR-UFD-00001
	PRO08-ING-FR01
	21044-PR-MSD-00001

P-332A/B LEAN AMINE PUMPS SPEED 2980 rpm RATED CAPACITY 70 m ³ /h HEAD 42 bar TYPE Multistage centrifugal MANUFACTURER Carbarino CASING DESIGN PRESSURE 55 barg CASING DESIGN TEMPERATURE -15/160 °C MODEL 65 GH 6 MOTOR POWER 160 kW CASING MATERIAL SS316L	MP-332A/B LEAN AMINE PUMP ELECTRIC MOTOR SPEED HOLD rpm RATED POWER 160 kW VOLTAGE 690 Vac FREQUENCY 50 Hz STARTING DOL ENCLOSURE PROTECTION IP55 STD IEC FRAME SIZE HOLD
F-331A/B PARTICULATE PRE FILTERS DESIGN PRESSURE 10 bar g DESIGN TEMPERATURE -15/160 °C DIAMETER 6" NPS SM TO SM 1100 mm CORROSION ALLOWANCE 0 mm INSULATION NO LINING/CLADDING N.A. MATERIAL SS304 INTERNAL VOLUME 0.023 m ³ INTERNAL ELEMENTS Cartridge (5 micron rating)	F-332A/B CHARCOAL FILTERS DESIGN PRESSURE 10 bar g DESIGN TEMPERATURE -15/160 °C DIAMETER 12" NPS SM TO SM 1000 mm CORROSION ALLOWANCE 0 mm INSULATION NO LINING/CLADDING N.A. MATERIAL SS304 INTERNAL VOLUME 0.07 m ³ INTERNAL ELEMENTS Charcoal + ceramic balls
F-333A/B PARTICULATE AFTER FILTERS DESIGN PRESSURE 10 bar g DESIGN TEMPERATURE -15/160 °C DIAMETER 6" NPS SM TO SM 1100 mm CORROSION ALLOWANCE 0 mm INSULATION NO LINING/CLADDING N.A. MATERIAL SS304 INTERNAL VOLUME 0.023 m ³ INTERNAL ELEMENTS Cartridge (5 micron rating)	P-337 ANTIFOAM MAKE UP PUMPS SPEED 1000 rpm RATED CAPACITY 0.18 m ³ /h HEAD 30 bar TYPE MEMBRANE (ELECTRIC) MANUFACTURER PROMINENT CASING DESIGN PRESSURE 10 barg CASING DESIGN TEMPERATURE -15/100 °C MODEL EXTRONIC G100 POWER SUPPLY 230vac/50Hz/1ph
E-334 LEAN AMINE COOLER DESIGN DUTY 2526 kW DESIGN PRES. 55 bar g DESIGN TEMP. -15/160 °C MATERIAL SS304 / A1 fin CORR. ALL. 0 mm FAN Qty 18 FAN ABS POWER each HOLD kW SURFACE AREA HOLD m ²	V-337 ANTIFOAM AGENT TANK DESIGN PRESSURE 50 mbar g DESIGN TEMPERATURE -15/47 °C DIAMETER 580 mm SM TO SM 880 mm CORROSION ALLOWANCE 0 mm INSULATION NO LINING/CLADDING N.A. MATERIAL Carbon steel INTERNAL VOLUME 0.2 m ³ INTERNAL ELEMENTS N/A
MF-334-1..18 LEAN AMINE COOLER FAN ELECTRIC MOTORS SPEED HOLD rpm RATED POWER HOLD kW VOLTAGE 400/3 Vac FREQUENCY 50 Hz STARTING DOL ENCLOSURE PROTECTION IP54 min STD IEC FRAME SIZE HOLD	F-334-1..18 FAN SPEED HOLD rpm RATED CAPACITY HOLD m ³ /h ABSORBED POWER HOLD kW

- NOTES:
- PUMPS CAN BE 2x100% or 3x50%
 - PCS PREVENTS TO RUN BOTH PUMPS AT THE SAME TIME. WHEN OPERATOR NEEDS TO CHANGE THE OPERATING PUMP FOR THE STAND BY PUMP, THE PUMPS WILL BE ALLOWED TO RUN SIMULTANEOUSLY FOR 5 MINUTES.
 - ALL INSTRUMENTS AND MANUAL VALVES ON THIS P&ID AND OUT OF SKID LIMITS WILL BE SUPPLIED BY TAM AS LOOSE ITEMS.
 - AMINE SAMPLING POINT.

ISA 0X	INSTRUMENT SUPPLY AIR
AOX 0X	AMINE DRAIN
AF 0X	ANTIFOAM
V 0X	VENT
----- TAM SUPPLY LIMIT	

6	27 Mar 2026	APPROVED FOR CONSTRUCTION	TAM (JB)	TAM (MM)	LOACPLONE
5	14 Oct 2025	APPROVED FOR CONSTRUCTION	TAM (LD)	LOACPLONE	W.BOUALAI
4	15 Feb 2024	APPROVED FOR CONSTRUCTION	TAM (LD)	LOACPLONE	W.BOUALAI
3	25 Jul 2023	ISSUED FOR APPROVAL	TAM (LD)	LOACPLONE	W.BOUALAI
2	06 Oct 2022	ISSUED FOR APPROVAL	TAM (LD)	LOACPLONE	W.BOUALAI
1	21 Jul 2022	ISSUED FOR APPROVAL	TAM (MT)	TAM (LD)	LOACPLONE
0	24 Jun 2022	ISSUED FOR APPROVAL	TAM (MT)	TAM (LD)	LOACPLONE
Rev.	Date	Description	Draft	Check'd	Approved
Customer: MANAENERGY					
Contractor: ITALFLUID					
Engineering: ITALFLUID					
Technology Vendor: TECHNICAL AMERICA INC					
OFFICES: Mahaj Rad Center, Batiment 6, Jeme etage, Avenue Al Az, Hay Ryad, 10100 Rabat, Morocco					
OFFICES: VIA VALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742					
OFFICES: VIA VALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742					
OFFICES: 301 N Smith Avenue, 92880 Corona, California (USA) Ph.+1 951 272 9540					
TITLE: LNG TENDRARA PLANT			OBJECT: P&ID - SWEETENING		
UNIT 330 SKID S-332/S-333					
TYPE	SCALE	DOCUMENTS ID	VENDOR DOC. CODE		SHEET/OF
A1	N/A	PRO13-ING-PRO7	21044-PR-PID-33006		1/1